

Java Stream API Employee Dataset - Answer Key

59 practice questions with expected outputs. Generated from the current app question data.

Filter	Q1. Find all employees whose salary is greater than 60,000.
Answer	John(101, 70000), Emma(105, 80000), Mike(106, 75000), Sophia(107, 90000), John(109, 70000)
Basic	Q2. Count the total number of employees.
Answer	10
Collector	Q3. Get a list of all employee names.
Answer	[John, Alice, Bob, David, Emma, Mike, Sophia, Chris, John, Robert]
Collector	Q4. Convert all employee names to uppercase.
Answer	[JOHN, ALICE, BOB, DAVID, EMMA, MIKE, SOPHIA, CHRIS, JOHN, ROBERT]
Sorting	Q5. Sort employees by salary in ascending order.
Answer	David(104,45000), Alice(102,50000), Chris(108,55000), Bob(103,60000), Robert(110,60000), John(101,70000), John(109,70000), Mike(106,75000), Emma(105,80000), Sophia(107,90000)
Sorting	Q6. Sort employees by salary in descending order.
Answer	Sophia(107,90000), Emma(105,80000), Mike(106,75000), John(101,70000), John(109,70000), Bob(103,60000), Robert(110,60000), Chris(108,55000), Alice(102,50000), David(104,45000)
Basic	Q7. Find the employee with the highest salary.
Answer	Sophia(107, 90000)
Basic	Q8. Find the employee with the lowest salary.
Answer	David(104, 45000)
Basic	Q9. Find the average salary of all employees.
Answer	65500.0
Basic	Q10. Find the total salary expenditure of the company.
Answer	655000
Filter	Q11. Find all employees belonging to the IT department.
Answer	John(101,70000), Bob(103,60000), Mike(106,75000), Robert(110,60000)
Filter	Q12. Find employees whose names start with 'J'.
Answer	John(101), John(109)
Filter	Q13. Find employees whose salary is between 50,000 and 80,000.

Answer	Alice(102,50000), Bob(103,60000), Chris(108,55000), Robert(110,60000), John(101,70000), John(109,70000), Mike(106,75000), Emma(105,80000)
Basic	Q14. Check whether all employees earn more than 40,000.
Answer	true
Basic	Q15. Check whether any employee belongs to Finance.
Answer	true
Basic	Q16. Check whether no employee has a salary less than 30,000.
Answer	true
Filter	Q17. Find the first employee whose salary is greater than 75,000.
Answer	Emma(105, 80000)
Filter	Q18. Find any employee from the HR department.
Answer	Alice(102) or Emma(105)
Grouping	Q19. Count employees in each department.
Answer	IT:4, HR:2, Admin:2, Finance:2
Grouping	Q20. Group employees by department.
Answer	IT:[John(101), Bob(103), Mike(106), Robert(110)]; HR:[Alice(102), Emma(105)]; Admin:[David(104), John(109)]; Finance:[Sophia(107), Chris(108)]
Grouping	Q21. Find the average salary department-wise.
Answer	IT:66250.0, HR:65000.0, Admin:57500.0, Finance:72500.0
Grouping	Q22. Find the highest-paid employee in each department.
Answer	IT: Mike(106,75000); HR: Emma(105,80000); Admin: John(109,70000); Finance: Sophia(107,90000)
Grouping	Q23. Find the lowest-paid employee in each department.
Answer	IT: Bob(103,60000) & Robert(110,60000); HR: Alice(102,50000); Admin: David(104,45000); Finance: Chris(108,55000)
Grouping	Q24. Find the department with the highest average salary.
Answer	Finance (72500.0)
Grouping	Q25. Partition employees based on salary > 70,000.
Answer	true: Emma(105,80000), Mike(106,75000), Sophia(107,90000); false: others
Grouping	Q26. Partition employees into IT and Non-IT groups.

Answer	IT:[John(101), Bob(103), Mike(106), Robert(110)]; Non-IT:[Alice(102), David(104), Emma(105), Sophia(107), Chris(108), John(109)]
Grouping	Q27. Group employees by salary range: Low <60k, Medium 60k-75k, High >75k.
Answer	Low:[David(104), Alice(102), Chris(108)]; Medium:[Bob(103), Robert(110), John(101), John(109), Mike(106)]; High:[Emma(105), Sophia(107)]
Collector	Q28. Find all distinct department names.
Answer	[IT, HR, Admin, Finance]
Collector	Q29. Find duplicate employee names.
Answer	John (appears twice)
Collector	Q30. Find duplicate salaries.
Answer	70000 (2), 60000 (2)
Collector	Q31. Find the frequency of employee names.
Answer	John:2; Alice:1; Bob:1; David:1; Emma:1; Mike:1; Sophia:1; Chris:1; Robert:1
Collector	Q32. Find the frequency of salaries.
Answer	70000:2; 60000:2; 45000:1; 50000:1; 80000:1; 75000:1; 90000:1; 55000:1
Grouping	Q33. Find employees having the same salary.
Answer	70000: John(101), John(109); 60000: Bob(103), Robert(110)
Grouping	Q34. Find departments having more than one employee.
Answer	IT, HR, Admin, Finance
Sorting	Q35. Find the second highest salary.
Answer	80000
Sorting	Q36. Find the third highest salary.
Answer	75000
Sorting	Q37. Find the Nth highest salary as distinct salaries descending.
Answer	[1:90000, 2:80000, 3:75000, 4:70000, 5:60000, 6:55000, 7:50000, 8:45000]
Sorting	Q38. Find the top 3 highest salaries.
Answer	[90000, 80000, 75000]
Sorting	Q39. Find the top 5 highest-paid employees.
Answer	Sophia(107,90000), Emma(105,80000), Mike(106,75000), John(101,70000), John(109,70000)

Sorting	Q40. Sort employees by department and then salary descending.
Answer	IT:[Mike(106,75000), John(101,70000), Bob(103,60000), Robert(110,60000)]; HR:[Emma(105,80000), Alice(102,50000)]; Admin:[John(109,70000), David(104,45000)]; Finance:[Sophia(107,90000), Chris(108,55000)]
Sorting	Q41. Sort employees by name and then salary.
Answer	[Alice(102,50000), Bob(103,60000), Chris(108,55000), David(104,45000), Emma(105,80000), John(101,70000), John(109,70000), Mike(106,75000), Robert(110,60000), Sophia(107,90000)]
Sorting	Q42. Find employees with the same salary sorted by name.
Answer	60000: [Bob(103), Robert(110)]; 70000: [John(101), John(109)]
Collector	Q43. Convert List<Employee> to List<String> containing names only.
Answer	[John, Alice, Bob, David, Emma, Mike, Sophia, Chris, John, Robert]
Collector	Q44. Convert List<Employee> to Map<Integer, Employee> using ID as key.
Answer	{101:John(IT,70000), 102:Alice(HR,50000), 103:Bob(IT,60000), 104:David(Admin,45000), 105:Emma(HR,80000), 106:Mike(IT,75000), 107:Sophia(Finance,90000), 108:Chris(Finance,55000), 109:John(Admin,70000), 110:Robert(IT,60000)}
Collector	Q45. Convert List<Employee> to Map<String, Double> where name maps to salary list because John repeats.
Answer	John:[70000,70000], Alice:[50000], Bob:[60000], David:[45000], Emma:[80000], Mike:[75000], Sophia:[90000], Chris:[55000], Robert:[60000]
Collector	Q46. Join all employee names into a comma-separated string.
Answer	John,Alice,Bob,David,Emma,Mike,Sophia,Chris,John,Robert
Collector	Q47. Join all distinct department names separated by " ".
Answer	IT HR Admin Finance
Collector	Q48. Create a map of Department -> Employee Count.
Answer	{IT:4, HR:2, Admin:2, Finance:2}
Collector	Q49. Create a map of Department -> Total Salary.
Answer	{IT:265000, HR:130000, Admin:115000, Finance:145000}
Advanced	Q66. Find employees whose salary is greater than their department average.
Answer	John(101, IT,70000), Mike(106, IT,75000), Emma(105, HR,80000), John(109, Admin,70000), Sophia(107, Finance,90000)
Basic	Q67. Find employees earning the maximum salary in the company.
Answer	Sophia(107, 90000)
Basic	Q68. Find employees earning the minimum salary in the company.

Answer	David(104, 45000)
Advanced	Q69. Find departments where average salary is greater than 60,000.
Answer	IT, HR, Finance
Grouping	Q70. Find the sum of salaries department-wise.
Answer	IT:265000, HR:130000, Admin:115000, Finance:145000
Advanced	Q71. Find the employee with the longest name.
Answer	Sophia(107) and Robert(110)
Advanced	Q72. Find the employee with the shortest name.
Answer	Bob(103)
Grouping	Q73. Find employees grouped by the first letter of their name.
Answer	A:[Alice(102)]; B:[Bob(103)]; C:[Chris(108)]; D:[David(104)]; E:[Emma(105)]; J:[John(101), John(109)]; M:[Mike(106)]; R:[Robert(110)]; S:[Sophia(107)]
Advanced	Q74. Create a custom collector to join employee names with " ".
Answer	John Alice Bob David Emma Mike Sophia Chris John Robert
Advanced	Q75. Create a custom collector to calculate total salary without using Collectors.summingDouble().
Answer	655000